

Pulkit Chawla

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SUMMARY

Software engineer (6+ years at Oracle Cloud and Qubole) turned AI engineer after an MSc in Data Science and AI at TU Eindhoven. In 2026 built production-grade agent systems: a verification kernel for agent-written code, a native macOS computer-use agent, and a personal email agent with structural prompt-injection defences. Owns services end to end: design, build, test, release, on-call.

Work authorisation: NL orientation-year permit (full labour-market access); HSM-eligible under the reduced graduate criterion

EXPERIENCE

Freelance and personal projects

Jan. 2026 – Present

Independent AI / software engineer

Eindhoven, Netherlands

- Designed and shipped four agent systems (below): a verification kernel for AI-written code, a native macOS computer-use agent, a personal email agent with structural prompt-injection defences, and a multi-account trading fan-out platform
- Built a WhatsApp AI assistant for an Eindhoven student food-catering business (Twilio, Gemini/Gemma, FastAPI on DigitalOcean) handling orders, meal selection, leave days and billing questions

Eindhoven University of Technology, IMAGEN Programme

Jan. 2025 – Nov. 2025

Graduate AI Researcher (MSc thesis), MLOps and Computer Vision

Eindhoven, Netherlands

- Built the MLOps platform behind IMAGEN, a Dutch government-funded animal-welfare research programme run by TU/e, Wageningen University and Hendrix Genetics
- Combined ClearML experiment tracking, SLURM job scheduling and PiCaS task distribution into a single workflow with fully reproducible runs in Singularity containers
- Wrote a FAIR data pipeline converting heterogeneous farm video datasets to HDF5 with embedded metadata, raising data-loading throughput and reducing GPU idle time during training
- Exposed the pipeline through a web UI and REST API so researchers could fine-tune detection models without logging into the HPC cluster; per-animal bounding boxes fed behavioural analysis such as feeding-order dominance

Oracle Cloud Infrastructure

Jan. 2021 – Aug. 2023

Member of Technical Staff, Big Data Service (BDS)

Bangalore, India

- Sole owner of the customer-facing console of OCI Big Data Service in a ~30-engineer team: built, tested, released and operated the Node.js UI used to manage Hadoop clusters, and trained other engineers to take it over before leaving
- Owned the Hive and HBase distributions: forked upstream open source, set up TeamCity pipelines that built and deployed every PR to staging automatically, and kept both patched by back-porting upstream CVE fixes, a hard requirement for enterprise customers leaving Cloudera
- Served in the production on-call rotation, resolving P0/P1 incidents with the US-based support and architecture teams

Qubole (acquired by Idera)

Jan. 2018 – Dec. 2020

Software Engineer → Software Engineer II, Cluster Orchestration

Bangalore, India

- One of three engineers owning the cluster-orchestration backend of Qubole's big-data platform, spinning up and managing Hadoop, Spark, Presto and Hive clusters inside customer accounts on AWS, Azure and OCI
- Added Datadog monitoring for cluster metrics and Jenkins QA pipelines gating every release on client-specific integration tests across staging and production
- Built secure tunnels between Qubole's control plane and clusters running in customers' own cloud accounts, letting customers operate their clusters directly from the UI

PROJECTS & RESEARCH

devteam: verification kernel for agent-written software | *Node.js*

Jul 2026 – present

- Every module is a contract (interface, tests, pinned toolchain) hashed into its identity; agent-written code is admitted only after passing in a network-less Docker container, run twice in two environments with transcripts diffed, plus held-out tests the author never sees
- Ran an end-to-end trial in which an AI manager hired its own team, drafted contracts from three examples and shipped gate-verified modules with zero human interventions; 356 commits, held-out vaults caught real bugs invisible to the visible suite
- Quarantined unverifiable effects (network, clocks, human input) into declared holes with owner-granted host allowlists judged on recorded real traffic; a headless real-mouse probe harness caught 15 bugs that unit tests had passed

- HumanHands: native macOS computer-use agent** | *Swift, SwiftUI*

2026
- Agent that perceives the screen through the macOS Accessibility tree (local, pixel-exact, no screenshots to the model), reasons with Claude, and drives the real cursor along human-like paths with human typing cadence; vision kept only as a concurrent fallback
 - Security spine: credentials live in the Keychain and reach the model only as placeholders substituted at the last instant; a domain-verified gate refuses to type a secret on an unconfirmed site; user-owned app and site allowlists bound its reach
 - Fully native Swift across 10 SwiftPM targets with every OS primitive behind a protocol boundary; 135 tests run off-device
- Personal email agent (NanoClaw fork)** | *TypeScript, Node.js, Claude Agent SDK*

2026
- Structural defence against prompt injection: a tool-less reader model is the only component that touches raw email, the acting agent sees only sanitised digests, and every outbound action stops at a one-tap approval that sends exactly the pre-created draft, enforced in host code rather than prompts
 - Extended the open-source NanoClaw runtime with an iMessage channel and a portable plain-file knowledge base (+16k lines); runs 24/7 on own hardware across three daemons (iMessage, ingestion, PWA) at zero idle cost
- LLM question answering with RAG (TU/e summer research)** | *Python, LangChain*

Jun – Aug 2024
- Built a retrieval-augmented question-answering pipeline over academic documents with LangChain, the OpenAI API and FAISS vector search
 - Benchmarked against BM25 and TF-IDF baselines and tuned chunking, context management and prompting (chain-of-thought, few-shot) to beat plain LLM answers on domain-specific queries
- trader-automata: multi-account order fan-out platform** | *Python, FastAPI*

2026
- Mirrors one trade decision across many brokerage accounts concurrently: pluggable broker layer on official SDKs, idempotent fan-out over an append-only event-sourced ledger, multi-tenant role-based auth, and Fernet-encrypted credentials with TOTP

EDUCATION

Eindhoven University of Technology (TU/e)	Aug. 2023 – Nov. 2025
<i>Master of Science in Data Science and Artificial Intelligence</i>	<i>Eindhoven, Netherlands</i>
• Thesis: MLOps and computer-vision pipeline for precision livestock farming (IMAGEN programme)	
Birla Institute of Technology, Mesra	Aug. 2014 – May 2018
<i>Bachelor of Engineering in Computer Science</i>	<i>Ranchi, India</i>

TECHNICAL SKILLS

AI/ML: LLM agents (Claude Agent SDK), RAG and vector search (FAISS), prompt-injection defences, LangChain, PyTorch, computer vision (object detection, tracking), fine-tuning, model serving and inference services, ClearML

Languages: Python, TypeScript/JavaScript, Swift, Java, SQL, Bash

Backend & Tooling: FastAPI, Spring, Node.js, REST APIs, Jenkins, GitHub Actions, CI/CD, Datadog, Elasticsearch, Playwright

Big Data & Cloud: Hadoop, Spark, AWS, Azure, OCI, Docker, Kubernetes

Spoken languages: English: fluent; Hindi: native; Dutch: beginner, in a course since Aug 2026